# Note 13.2 Integrals of Vector Functions; Projectile Motion

# Integrals of Vector Functions

## Definition of Indefinite Integral

The indefinite integral of $\mathbf{r}$ with respect to $t$ is the set of all antiderivatives of $\mathbf{r}$, denoted by $\int \mathbf{r}(t)\, dt$. If $\mathbf{R}$ is any antiderivative of $\mathbf{r}$, then

$$\int \mathbf{r}(t)\, dt = \mathbf{R}(t) + \mathbf{C}$$

For the components of $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, we can write that

$$\int \mathbf{r}(t)\, dt = \left(\int f(t)\, dt\right)\mathbf{i} + \left(\int g(t)\, dt\right)\mathbf{j} + \left(\int h(t)\, dt\right)\mathbf{k}$$

## Definition of Definite Integral

For the components of $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ integrable over $[a, b]$, then the definite integral of $\mathbf{r}$ from $a$ to $b$ is

$$\int_a^b \mathbf{r}(t)\, dt = \left(\int_a^b f(t)\, dt\right)\mathbf{i} + \left(\int_a^b g(t)\, dt\right)\mathbf{j} + \left(\int_a^b h(t)\, dt\right)\mathbf{k}$$

# Exercises

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